

Total No. of Questions : 6]

SEAT No. :

P5045

[Total No. of Pages : 2

T.E./Insem. - 543

T.E. (I.T.) (Semester - I)

DATABASE MANAGEMENT SYSTEMS

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4. Q5 or Q.6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) What is data independency? Why it is lacking in file processing system.[3]

b) Explain following terms with examples. [3]

Superkey, candidate key, primary key

c) Consider a relation professor. [4]

Professor = {E_no, E_name, Prof_Rank, Prof_office, D_code, D_name, Advisee, Committee_code, Journal_code}

Professor can have multiple advisee, can work on multiple committees, can edit more than on journals.

- 1) Draw an dependency diagram.
- 2) Identified multivalued attribute.
- 3) Redraw an dependency diagram to yield a set of 3NF
- 4) Eliminate multivalued dependency by converting the table in 4NF.

OR

Q2) a) Consider a relation schema R(ABCD) with FDs [4]

$AB \rightarrow C$, $AB \rightarrow D$, $C \rightarrow A$ Show that schema is in 3NF but not in BCNF.

b) Draw an ER Diagram for wholesale dealers for audio, video consumer equipment from different manufacturing (BRAND), Customers are the various retail outlets (RETAILERS), wholesalers extends credit to OLD customers and special discounts are offered to NEW customers. (retailers), ER Scope is restricted to details (queries on) of customers (retailers), products, stocks, and their discounts credit offers etc. [6]

P.T.O.

Q3) a) Consider following relations [5]

Customer = {Cid, Cname, city, State}

Order = {Oid, Cid, Odate, amount}

- 1) List all customers who belongs to 'Maharashtra' state and order by city
 - 2) What are the name and amount of all customers who placed orders between date '1-1-2011 to 31-12-2014'
 - 3) Write DDL command to add constraint on order table to check amount is always > 0 and cid is a foreign key.
- b) What is cursor? Need for it? Explain different phases of cursors with example. [5]

OR

Q4) a) Write short note on (Any two) [6]

- i) Null values in SQL (Aggregation, AND, OR, NOT operators)
 - ii) Correlated Queries
 - iii) Triggers in SQL
- b) Write down difference between TRUNCATE and DELETE with example. What is advantage of TRUNCATE over delete and vice versa. [4]

Q5) a) Rewrite the following schedule using Two Phase Locking protocol and state if it is conflict serializable or not using precedence graph. [6]

T1

Read (A)

Write (A)

Read (B)

Write (B)

T2

Read (A)

Write (A)

Read (B)

Write (B)

- b) What is recoverable schedule? Why it is desirable? [4]

OR

Q6) a) What is deadlock? Give example? Discuss deadlock prevention techniques. [5]

- b) State the goal of concurrency control system? How it is achieved with timestamp based protocol? [5]

